1) a)

$r = 4\cos\theta \Rightarrow r\sin\theta = 4\cos\theta\sin\theta$ so $y = 4\cos\theta\sin\theta$

$r = 4\cos\theta \Rightarrow r\cos\theta = 4\cos^2\theta$ so $x = 4\cos^2\theta$

$\therefore y = 4\cos\theta\sin\theta$ and $x = 4\cos^2\theta$

1) b)

$r = 8\sin\theta \Rightarrow r\sin\theta = 8\sin^2\theta$ so $y = 8\sin^2\theta$

$r = 8\sin\theta \Rightarrow r\cos\theta = 8\sin\theta\cos\theta$ so $x = 8\sin\theta\cos\theta$

$\therefore y = 8\sin^2\theta$ and $x = 8\sin\theta\cos\theta$

1) c)

$r = 3 + 3\cos\theta \Rightarrow r\sin\theta = 3\sin\theta + 3\sin\theta\cos\theta$ so $y = 3\sin\theta + 3\sin\theta\cos\theta$

$r = 3 + 3\cos\theta \Rightarrow r\cos\theta = 3\cos\theta + 3\cos^2\theta$ so $x = 3\cos\theta + 3\cos^2\theta$

$\therefore y = 3\sin\theta + 3\sin\theta\cos\theta$ and $x = 3\cos\theta + 3\cos^2\theta$

1) d)

$r = 6 - 6\sin\theta \Rightarrow r\sin\theta = 6\sin\theta - 6\sin^2\theta$ so $y = 6\sin\theta - 6\sin^2\theta$

$r = 6 - 6\sin\theta \Rightarrow r\cos\theta = 6\cos\theta - 6\sin\theta\cos\theta$ so $x = 6\cos\theta - 6\sin\theta\cos\theta$

$\therefore y = 6\sin\theta - 6\sin^2\theta$ and $x = 6\cos\theta - 6\sin\theta\cos\theta$

1) e)

$r = 5\sin 2\theta \Rightarrow r\sin\theta = 5\sin 2\theta\sin\theta$ so $y = 10\cos\theta\sin^2\theta$

$r = 5\sin 2\theta \Rightarrow r\cos\theta = 5\sin 2\theta\cos\theta$ so $x = 10\sin\theta\cos^2\theta$

$\therefore y = 10\cos\theta\sin^2\theta$ and $x = 10\sin\theta\cos^2\theta$

2) a)

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = -3\sin\theta - 6\cos\theta\sin\theta$

2) b)

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = 5\cos\theta - 10\sin\theta\cos\theta$

2) c)

$\frac{dy}{d\theta} = -7\sin\theta + 14\cos\theta\sin\theta$

2) d)

$\frac{dy}{d\theta} = -2\cos\theta - 4\sin\theta\cos\theta$

2) e)

$\frac{dy}{d\theta} = -24\cos^2\theta\sin\theta + 4\sin\theta$

2) f)

$\frac{dy}{d\theta} = -6\sin^3\theta + 12\sin\theta\cos^2\theta$

3) a)

$r = 2 + 2\sin\theta \Rightarrow r\sin\theta = 2\sin\theta + 2\sin^2\theta$

$y = 2\sin\theta + 2\sin^2\theta$

$\frac{dy}{d\theta} = 2\cos\theta + 4\sin\theta\cos\theta = 2\cos\theta + 2\sin 2\theta$

3) b)

$r = 2 + 2\sin\theta \Rightarrow r\cos\theta = 2\cos\theta + 2\sin\theta\cos\theta$

$x = 2\cos\theta + \sin 2\theta$

$\frac{dx}{d\theta} = -2\sin\theta + 2\cos 2\theta$

3) c)

$2\cos\theta + 2\sin 2\theta = 2\cos\theta + 4\sin\theta\cos\theta = 2\cos\theta(1 + 2\sin\theta) = 0$

So $\cos\theta = 0$ or $\sin\theta = -\frac{1}{2}$

$\therefore \theta = -\frac{\pi}{2}, \theta = \frac{\pi}{2}, \theta = -\frac{\pi}{6}$ or $\theta = -\frac{5\pi}{6}$

3) d)

$-2\sin\theta + 2\cos 2\theta = -2\sin\theta + (1 - 2\sin^2\theta) = 0$

So $2\sin^2\theta + \sin\theta - 1 = (2\sin\theta - 1)(\sin\theta + 1) = 0$

So $\sin\theta = \frac{1}{2}$ or $\sin\theta = -1$

$\therefore \theta = \frac{\pi}{6}$, $\theta = \frac{5\pi}{6}$ or $\theta = -\frac{\pi}{2}$

3) e)

When $\theta = -\frac{\pi}{2}$ both $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ so this is the value of θ for which there is a cusp point.

4) a)

Tangent parallel to the initial line $\theta = 0$: $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0$

$r = 2 + 2\cos\theta \Rightarrow r\sin\theta = 2\sin\theta + 2\sin\theta\cos\theta \Rightarrow y = 2\sin\theta + 2\sin\theta\cos\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = 2\cos\theta - 2\sin^2\theta + 2\cos^2\theta$

$r = 2 + 2\cos\theta \Rightarrow r\cos\theta = 2\cos\theta + 2\cos^2\theta \Rightarrow x = 2\cos\theta + 2\cos^2\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -2\sin\theta - 4\cos\theta\sin\theta$

Consider $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$:

$2\cos\theta - 2(1 - \cos^2\theta) + 2\cos^2\theta = 0$

$2\cos\theta - 2 + 2\cos^2\theta + 2\cos^2\theta = 0$

$4\cos^2\theta + 2\cos\theta - 2 = 0$

$2\cos^2\theta + \cos\theta - 1 = 0$

$(\cos\theta + 1)(2\cos\theta - 1) = 0$

So $\cos\theta = -1$ or $\cos\theta = \frac{1}{2}$

$\therefore \theta = \pi$, $\theta = \frac{\pi}{3}$ or $\theta = -\frac{\pi}{3}$

When $\theta = \pi$, $y = 2\sin\pi + 2\sin\pi\cos\pi = 0$ but $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ so a cusp point.

When $\theta = \frac{\pi}{3}$, $y = 2\sin\frac{\pi}{3} + 2\sin\frac{\pi}{3}\cos\frac{\pi}{3} = \frac{3\sqrt{3}}{2}$

When $\theta = -\frac{\pi}{3}$, $y = 2\sin\left(-\frac{\pi}{3}\right) + 2\sin\left(-\frac{\pi}{3}\right)\cos\left(-\frac{\pi}{3}\right) = -\frac{3\sqrt{3}}{2}$

So the tangent lines are $y = \sqrt{3} + \frac{\sqrt{3}}{2}$ and $y = -\frac{3\sqrt{3}}{2}$

4) b)

Tangent parallel to the initial line $\theta = 0$: $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0$

$r = 3 - 3\sin\theta \Rightarrow r\sin\theta = 3\sin\theta - 3\sin^2\theta \Rightarrow y = 3\sin\theta - 3\sin^2\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = 3\cos\theta - 6\cos\theta\sin\theta$

$r = 3 - 3\sin\theta \Rightarrow r\cos\theta = 3\cos\theta - 3\sin\theta\cos\theta \Rightarrow x = 3\cos\theta - 3\sin\theta\cos\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -3\sin\theta + 3\sin^2\theta - 3\cos^2\theta$

Consider $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$:

$3\cos\theta\,(1 - 2\sin\theta) = 0$

So $\cos\theta = 0$ or $\sin\theta = \frac{1}{2}$

$\therefore \theta = \frac{\pi}{2}, \theta = -\frac{\pi}{2}, \theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$

When $\theta = \frac{\pi}{2}$, $y = 3\sin\frac{\pi}{2} - 3\sin^2\left(\frac{\pi}{2}\right) = 0$ but $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ so a cusp point.

When $\theta = -\frac{\pi}{2}$, $y = 3\sin\left(-\frac{\pi}{2}\right) - 3\sin^2\left(-\frac{\pi}{2}\right) = -6$

When $\theta = \frac{\pi}{6}$, $y = 3\sin\frac{\pi}{6} - 3\sin^2\left(\frac{\pi}{6}\right) = \frac{3}{4}$

When $\theta = \frac{5\pi}{6}$, $y = 3\sin\frac{5\pi}{6} - 3\sin^2\left(\frac{5\pi}{6}\right) = \frac{3}{4}$

So the tangent lines are $y = -6$ and $y = \frac{3}{4}$

4) c)

Tangent parallel to the initial line $\theta = 0$: $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0$

$r = 4\sin 2\theta \Rightarrow r\sin\theta = 4\sin\theta\sin 2\theta \Rightarrow y = 4\sin\theta\sin 2\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = 8\sin\theta\cos 2\theta + 4\cos\theta\sin 2\theta$

$r = 4\sin 2\theta \Rightarrow r\cos\theta = 4\sin 2\theta\cos\theta \Rightarrow x = 4\sin 2\theta\cos\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -4\sin 2\theta\sin\theta + 8\cos 2\theta\cos\theta$

Consider $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$:

$8\sin\theta\cos 2\theta + 4\cos\theta\sin 2\theta = 8\sin\theta(\cos^2\theta - \sin^2\theta) + 8\cos^2\theta\sin\theta$

$= 8\sin\theta(\cos^2\theta - \sin^2\theta + \cos^2\theta)$

$= 8\sin\theta(2\cos^2\theta - \sin^2\theta)$

$8\sin\theta(2\cos^2\theta - \sin^2\theta) = 0$

So $\sin\theta = 0$ or $2\cos^2\theta - \sin^2\theta = 0 \Rightarrow \tan^2\theta = 2$

$\therefore \theta = 0, \theta = \pi, \theta = -\frac{\pi}{2}$, $\tan\theta = \sqrt{2}$ or $\tan\theta = -\sqrt{2}$

When $\theta = 0, \pi$ $\quad y = 0$ and $\frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0$

When $\tan\theta = \sqrt{2}$ then $\sin\theta = \sqrt{\frac{2}{3}}$ and $\cos\theta = \frac{1}{\sqrt{3}}$,

so $\quad y = 8\sin^2\theta\cos\theta = 8 \times \frac{2}{3} \times \frac{1}{\sqrt{3}} = \frac{16\sqrt{3}}{9}$

When $\tan\theta = -\sqrt{2}$ then $\sin\theta = \sqrt{\frac{2}{3}}$ and $\cos\theta = -\frac{1}{\sqrt{3}}$,

so $\quad y = 8\sin^2\theta\cos\theta = 8 \times \frac{2}{3} \times -\frac{1}{\sqrt{3}} = -\frac{16\sqrt{3}}{9}$

So the tangent lines are $y = 0$, $y = \frac{16\sqrt{3}}{9}$ and $y = -\frac{16\sqrt{3}}{9}$

4) d)

Tangent parallel to the initial line $\theta = 0$: $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0$

$r = a + a\sin\theta \Rightarrow r\sin\theta = a\sin\theta + a\sin^2\theta \Rightarrow y = a\sin\theta + a\sin^2\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = a\cos\theta + 2a\sin\theta\cos\theta$

$r = a + a\sin\theta \Rightarrow r\cos\theta = a\cos\theta + a\sin\theta\cos\theta \Rightarrow x = a\cos\theta + \frac{a}{2}\sin 2\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -a\sin\theta + a\cos 2\theta$

Consider $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$:

$a\cos\theta + 2a\sin\theta\cos\theta = a\cos\theta\,(1 + 2\sin\theta) = 0$

So $\cos\theta = 0$ or $\sin\theta = -\frac{1}{2}$

$\therefore \theta = -\frac{\pi}{2}, \theta = \frac{\pi}{2}, \theta = -\frac{\pi}{6}, \theta = -\frac{5\pi}{6}$

When $\theta = -\frac{\pi}{2}$, $y = 0$ and but $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ so a cusp point.

When $\theta = \frac{\pi}{2}$, $y = 2a$

When $\theta = -\frac{\pi}{6}$, $y = -\frac{1}{4}a$

When $\theta = -\frac{5\pi}{6}$, $y = -\frac{1}{4}a$

So the tangent lines are $y = 2a$ and $y = -\frac{1}{4}a$

4) e)

Tangent parallel to the initial line $\theta = 0$: $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0$

$r = a - a\cos\theta \Rightarrow r\sin\theta = a\sin\theta - a\sin\theta\cos\theta \Rightarrow y = a\sin\theta - \frac{a}{2}\sin 2\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = a\cos\theta - a\cos 2\theta$

$r = a - a\cos\theta \Rightarrow r\cos\theta = a\cos\theta - a\cos^2\theta \Rightarrow x = a\cos\theta - a\cos^2\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -a\sin\theta + 2a\cos\theta\sin\theta$

Consider $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$:

$a\cos\theta - a\cos 2\theta = a\cos\theta - a(2\cos^2\theta - 1) = a\cos\theta - 2a\cos^2\theta + a$

$\therefore 2a\cos^2\theta - a\cos\theta - a = a(\cos\theta - 1)(2\cos\theta + 1) = 0$

So $\cos\theta = 1$ or $\cos\theta = -\frac{1}{2}$

$\therefore \theta = 0, \theta = -\frac{2\pi}{3}, \theta = \frac{\pi}{3}$

When $\theta = 0$, $y = 0$ but $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ so a cusp point.

When $\theta = -\frac{2\pi}{3}$, $y = -\frac{3\sqrt{3}}{4}a$

When $\theta = \frac{2\pi}{3}$, $y = \frac{3\sqrt{3}}{4}a$

So the tangent lines are $y = -\frac{3\sqrt{3}}{4}a$ and $y = \frac{3\sqrt{3}}{4}a$

4) f)

Tangent parallel to the initial line $\theta = 0$: $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0$

$r = a\cos 2\theta \Rightarrow r\sin\theta = a\sin\theta\cos 2\theta = a\sin\theta - 2a\sin^3\theta \Rightarrow y = a\sin\theta - 2a\sin^3\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = a\cos\theta - 6a\sin^2\theta\cos\theta$

$r = a\cos 2\theta \Rightarrow r\cos\theta = a\cos\theta\cos 2\theta = a\cos\theta\,(2\cos^2\theta - 1)$

$\Rightarrow x = 2a\cos^3\theta - a\cos\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -6a\cos^2\theta\sin\theta + a\sin\theta$

Consider $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$:

$a\cos\theta - 6a\sin^2\theta\cos\theta = a\cos\theta\,(1 - 6\sin^2\theta)$

$\therefore a\cos\theta\,(1 - 6\sin^2\theta) = 0$

So $\cos\theta = 0$ or $\sin\theta = \pm\frac{1}{\sqrt{6}}$

$\therefore \theta = -\frac{\pi}{2}, \theta = \frac{\pi}{2}$, $\sin\theta = \frac{1}{\sqrt{6}}$ or $\sin\theta = -\frac{1}{\sqrt{6}}$

When $\theta = -\frac{\pi}{2}$, $y = a$

When $\theta = \frac{\pi}{2}$, $y = -a$

When $\sin\theta = \frac{1}{\sqrt{6}}$, $y = \frac{1}{\sqrt{6}}a - 2a\left(\frac{1}{\sqrt{6}}\right)^3 = \frac{\sqrt{6}}{9}a$

When $\sin\theta = -\frac{1}{\sqrt{6}}$, $y = -\frac{1}{\sqrt{6}}a - 2a\left(-\frac{1}{\sqrt{6}}\right)^3 = -\frac{\sqrt{6}}{9}a$

So the tangent lines are $y = a$, $y = -a$, $y = \frac{\sqrt{6}}{9}a$ and $y = -\frac{\sqrt{6}}{9}a$

4) g)

Tangent parallel to the initial line $\theta = 0$: $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0$

$r = \mathrm{e}^{\theta} \Rightarrow r\sin\theta = \mathrm{e}^{\theta}\sin\theta \Rightarrow y = \mathrm{e}^{\theta}\sin\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = \mathrm{e}^{\theta}\cos\theta + \mathrm{e}^{\theta}\sin\theta$

$r = \mathrm{e}^{\theta} \Rightarrow r\cos\theta = \mathrm{e}^{\theta}\cos\theta \Rightarrow x = \mathrm{e}^{\theta}\cos\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -\mathrm{e}^{\theta}\sin\theta + \mathrm{e}^{\theta}\cos\theta$

Consider $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$:

$\mathrm{e}^{\theta}\cos\theta + \mathrm{e}^{\theta}\sin\theta = \mathrm{e}^{\theta}(\cos\theta + \sin\theta)$

$\mathrm{e}^{\theta}(\cos\theta + \sin\theta) = 0$

$\cos\theta + \sin\theta = 0 \Rightarrow \sin\theta = -\cos\theta$

So $\tan\theta = -1$

$\therefore \theta = -\frac{\pi}{4}$ or $\theta = \frac{3\pi}{4}$

When $\theta = -\frac{\pi}{4}$, $\quad y = -\frac{\sqrt{2}}{2}\mathrm{e}^{-\frac{\pi}{4}}$

When $\theta = \frac{3\pi}{4}$, $\quad y = \frac{\sqrt{2}}{2}\mathrm{e}^{\frac{3\pi}{4}}$

So the tangent lines are $y = -\frac{\sqrt{2}}{2}\mathrm{e}^{-\frac{\pi}{4}}$ and $y = \frac{\sqrt{2}}{2}\mathrm{e}^{\frac{3\pi}{4}}$

5) a)

Tangent perpendicular to the initial line $\theta = 0$: $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}y}{\mathrm{d}\theta} \neq 0$

$r = 4 - 4\cos\theta \Rightarrow r\cos\theta = 4\cos\theta - 4\cos^2\theta \Rightarrow x = 4\cos\theta - 4\cos^2\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -4\sin\theta + 8\cos\theta\sin\theta$

$r = 4 - 4\cos\theta \Rightarrow r\sin\theta = 4\sin\theta - 4\cos\theta\sin\theta \Rightarrow y = 4\sin\theta - 2\sin 2\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = 4\cos\theta - 4\sin 2\theta$

Consider $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$:

$-4\sin\theta + 8\cos\theta\sin\theta$

$4\sin\theta(2\cos\theta - 1) = 0$

So $\sin\theta = 0$ or $\cos\theta = \frac{1}{2}$

$\therefore \theta = 0, \theta = \pi, \theta = -\frac{\pi}{3}$ or $\theta = \frac{\pi}{3}$

When $\theta = 0$, $x = 0$ but $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ so a cusp point.

When $\theta = \pi$, $x = -8$

When $\theta = -\frac{\pi}{3}$, $x = 1$

When $\theta = \frac{\pi}{3}$, $x = 1$

So the tangent lines are $x = -8$ and $x = 1$

5) b)

Tangent perpendicular to the initial line $\theta = 0$: $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}y}{\mathrm{d}\theta} \neq 0$

$r = 2 + 2\sin\theta \Rightarrow r\cos\theta = 2\cos\theta + 2\sin\theta\cos\theta \Rightarrow x = 2\cos\theta + \sin 2\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -2\sin\theta + 2\cos 2\theta$

$r = 2 + 2\sin\theta \Rightarrow r\sin\theta = 2\sin\theta + 2\sin^2\theta \Rightarrow y = 2\sin\theta + 2\sin^2\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = 2\cos\theta + 4\sin\theta\cos\theta$

Consider $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$:

$-2\sin\theta + 2\cos 2\theta$

$= -2\sin\theta + 2(1 - 2\sin^2\theta)$

$= -2\sin\theta + 2 - 4\sin^2\theta$

$-2(2\sin\theta - 1)(\sin\theta + 1) = 0$

So $\sin\theta = \frac{1}{2}$ or $\sin\theta = -1$

$\therefore \theta = \frac{\pi}{6}$, $\theta = \frac{5\pi}{6}$, or $\theta = -\frac{\pi}{2}$

When $\theta = \frac{\pi}{6}$, $x = \frac{3\sqrt{3}}{2}$

When $\theta = \frac{5\pi}{6}$, $x = -\frac{3\sqrt{3}}{2}$

When $\theta = -\frac{\pi}{2}$, $x = 0$ but $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ so a cusp point.

So the tangent lines are $x = \frac{3\sqrt{3}}{2}$ and $x = -\frac{3\sqrt{3}}{2}$

5) c)

Tangent perpendicular to the initial line $\theta = 0$: $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}y}{\mathrm{d}\theta} \neq 0$

$r = 2\cos 2\theta \Rightarrow r\cos\theta = 2\cos\theta\cos 2\theta \Rightarrow x = 2\cos\theta\cos 2\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -4\cos\theta\sin 2\theta - 2\sin\theta\cos 2\theta$

$r = 2\cos 2\theta \Rightarrow r\sin\theta = 2\sin\theta\cos 2\theta \Rightarrow y = 2\sin\theta\cos 2\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = -4\sin\theta\sin 2\theta + 2\cos\theta\cos 2\theta$

Consider $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$:

$-4\cos\theta\sin 2\theta - 2\sin\theta\cos 2\theta$

$= -8\cos^2\theta\sin\theta - 2\sin\theta(2\cos^2\theta - 1)$

$= -2\sin\theta(4\cos^2\theta + 2\cos^2\theta - 1)$

$-2\sin\theta(6\cos^2\theta - 1) = 0$

So $\sin\theta = 0$ or $\cos\theta = \pm\frac{1}{\sqrt{6}}$

$\therefore \theta = 0, \theta = \pi, \cos\theta = \frac{1}{\sqrt{6}}$ or $\cos\theta = -\frac{1}{\sqrt{6}}$

When $\theta = 0$ $\quad x = 2$

When $\theta = \pi$ $\quad x = -2$

When $\cos\theta = \frac{1}{\sqrt{6}}$ then $\sin\theta = \frac{\sqrt{30}}{6}$,

so $\quad y = 2\cos\theta(\cos^2\theta - \sin^2\theta) = 2 \times \frac{1}{\sqrt{6}}\left(\frac{1}{6} - \frac{30}{36}\right) = -\frac{2\sqrt{6}}{9}$

When $\cos\theta = -\frac{1}{\sqrt{6}}$ then $\sin\theta = -\frac{\sqrt{30}}{6}$,

so $\quad y = 2\cos\theta(\cos^2\theta - \sin^2\theta) = 2 \times -\frac{1}{\sqrt{6}}\left(\frac{1}{6} - \frac{30}{36}\right) = \frac{2\sqrt{6}}{9}$

So the tangent lines are $x = -2$, $x = 2$, $x = -\frac{2\sqrt{6}}{9}$ and $x = \frac{2\sqrt{6}}{9}$

5) d)

Tangent perpendicular to the initial line $\theta = 0$: $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}y}{\mathrm{d}\theta} \neq 0$

$r = a - a\sin\theta \Rightarrow r\cos\theta = a\cos\theta - a\sin\theta\cos\theta \Rightarrow x = a\cos\theta - \frac{a}{2}\sin 2\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -a\sin\theta + a\cos 2\theta$

$r = a - a\sin\theta \Rightarrow r\sin\theta = a\sin\theta - a\sin^2\theta \Rightarrow y = a\sin\theta - a\sin^2\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = a\cos\theta + 2a\sin\theta\cos\theta$

Consider $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$:

$-a\sin\theta + a\cos 2\theta$

$= -a\sin\theta + 2a(1 - 2\sin^2\theta)$

$= -a\sin\theta + 2a - 2a\sin^2\theta$

$-a(2\sin\theta + 1)(\sin\theta - 1) = 0$

So $\sin\theta = -\frac{1}{2}$ or $\sin\theta = 1$

$\therefore \theta = \frac{\pi}{6}$, $\theta = \frac{5\pi}{6}$, or $\theta = \frac{\pi}{2}$

When $\theta = -\frac{\pi}{6}$, $x = \frac{3\sqrt{3}}{4}a$

When $\theta = -\frac{5\pi}{6}$, $x = -\frac{3\sqrt{3}}{4}a$

When $\theta = \frac{\pi}{2}$, $x = 0$ but $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ so a cusp point.

So the tangent lines are $x = \frac{3\sqrt{3}}{4}a$ and $x = -\frac{3\sqrt{3}}{4}a$

5) e)

Tangent perpendicular to the initial line $\theta = 0$: $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}y}{\mathrm{d}\theta} \neq 0$

$r = a + a\cos\theta \Rightarrow r\cos\theta = a\cos\theta + a\cos^2\theta \Rightarrow x = a\cos\theta + a\cos^2\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -a\sin\theta - 2a\cos\theta\sin\theta$

$r = a + a\cos\theta \Rightarrow r\sin\theta = a\sin\theta + a\cos\theta\sin\theta \Rightarrow y = a\sin\theta + \frac{a}{2}\sin 2\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = a\cos\theta + a\cos 2\theta$

Consider $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$:

$-a\sin\theta - 2a\cos\theta\sin\theta$

$-a\sin\theta\,(2\cos\theta + 1) = 0$

So $\sin\theta = 0$ or $\cos\theta = -\frac{1}{2}$

$\therefore \theta = 0, \theta = \pi, \theta = -\frac{2\pi}{3}$ or $\theta = \frac{2\pi}{3}$

When $\theta = 0$, $x = 2a$

When $\theta = \pi$, $x = 0$ but $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ so a cusp point.

When $\theta = -\frac{2\pi}{3}$, $x = -\frac{1}{4}a$

When $\theta = \frac{2\pi}{3}$, $x = -\frac{1}{4}a$

So the tangent lines are $x = 2a$ and $x = -\frac{1}{4}a$

5) f)

Tangent perpendicular to the initial line $\theta = 0$: $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}y}{\mathrm{d}\theta} \neq 0$

$r = a\sin 2\theta \Rightarrow r\cos\theta = a\cos\theta\sin 2\theta = 2a\cos^2\theta\sin\theta \Rightarrow x = 2a\cos^2\theta\sin\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = 2a\cos^3\theta - 4a\cos\theta\sin^2\theta$

$r = a\cos 2\theta \Rightarrow r\sin\theta = a\sin\theta\sin 2\theta = 2a\sin^2\theta\cos\theta \Rightarrow y = 2a\sin^2\theta\cos\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = -2a\sin^3\theta + 4a\sin\theta\cos^2\theta$

Consider $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$:

$2a\cos^3\theta - 4a\cos\theta\sin^2\theta$

$2a\cos\theta(\cos^2\theta - 2\sin^2\theta)$

$2a\cos\theta(1 - \sin^2\theta - 2\sin^2\theta)$

$-2a\cos\theta(1 - 3\sin^2\theta) = 0$

So $\cos\theta = 0$ or $\sin\theta = \pm\frac{1}{\sqrt{3}}$

$\therefore \theta = -\frac{\pi}{2}$, $\sin\theta = \frac{1}{\sqrt{3}}$ or $\sin\theta = -\frac{1}{\sqrt{3}}$

When $\theta = -\frac{\pi}{2}$, $x = 0$ but $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ so a cusp point.

When $\sin\theta = \frac{1}{\sqrt{3}}$ then $\cos\theta = \frac{\sqrt{6}}{3}$,

so $x = 2a\cos^2\theta\sin\theta = 2a \times \frac{6}{9} \times \frac{1}{\sqrt{3}} = \frac{4\sqrt{3}}{9}a$

When $\sin\theta = -\frac{1}{\sqrt{3}}$ then $\cos\theta = -\frac{\sqrt{6}}{3}$,

so $x = 2a\cos^2\theta\sin\theta = 2a \times \frac{6}{9} \times -\frac{1}{\sqrt{3}} = -\frac{4\sqrt{3}}{9}a$

So the tangent lines are $x = \frac{4\sqrt{3}}{9}a$ and $x = -\frac{4\sqrt{3}}{9}a$

5) g)

Tangent perpendicular to the initial line $\theta = 0$: $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}y}{\mathrm{d}\theta} \neq 0$

$r = \mathrm{e}^{\theta} \Rightarrow r\cos\theta = \mathrm{e}^{\theta}\cos\theta \Rightarrow x = \mathrm{e}^{\theta}\cos\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \mathrm{e}^{\theta}\cos\theta - \mathrm{e}^{\theta}\sin\theta$

$r = \mathrm{e}^{\theta} \Rightarrow r\sin\theta = \mathrm{e}^{\theta}\sin\theta \Rightarrow y = \mathrm{e}^{\theta}\sin\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = \mathrm{e}^{\theta}\sin\theta + \mathrm{e}^{\theta}\cos\theta$

Consider $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$:

$\mathrm{e}^{\theta}\cos\theta - \mathrm{e}^{\theta}\sin\theta$

$\mathrm{e}^{\theta}\cos\theta - \mathrm{e}^{\theta}\sin\theta = \mathrm{e}^{\theta}(\cos\theta - \sin\theta)$

$\mathrm{e}^{\theta}(\cos\theta - \sin\theta) = 0$

$\cos\theta - \sin\theta = 0 \Rightarrow \sin\theta = \cos\theta$

So $\tan\theta = 1$

$\therefore \theta = \frac{\pi}{4}$ or $\theta = -\frac{3\pi}{4}$

When $\theta = \frac{\pi}{4}$, $\quad x = \frac{\sqrt{2}}{2}\mathrm{e}^{\frac{\pi}{4}}$

When $\theta = -\frac{3\pi}{4}$, $\quad x = -\frac{\sqrt{2}}{2}\mathrm{e}^{-\frac{3\pi}{4}}$

So the tangent lines are $x = \frac{\sqrt{2}}{2}\mathrm{e}^{\frac{\pi}{4}}$ and $x = -\frac{\sqrt{2}}{2}\mathrm{e}^{-\frac{3\pi}{4}}$

5) h)

Tangent perpendicular to the initial line $\theta = 0$: $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}y}{\mathrm{d}\theta} \neq 0$

$r = 1 + \tan\theta \Rightarrow r\cos\theta = \cos\theta + \cos\theta\tan\theta \Rightarrow x = \cos\theta + \cos\theta\tan\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -\sin\theta + \cos\theta\sec^2\theta - \sin\theta\tan\theta$

$r = 1 + \tan\theta \Rightarrow r\sin\theta = \sin\theta + \sin\theta\tan\theta \Rightarrow y = \sin\theta + \sin\theta\tan\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = \cos\theta + \sin\theta\sec^2\theta + \cos\theta\tan\theta$

Consider $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$:

$-\sin\theta + \cos\theta\sec^2\theta - \sin\theta\tan\theta = 0$

So $-\sin\theta + \frac{1}{\cos\theta} - \frac{\sin^2\theta}{\cos\theta} = 0$

$\Rightarrow -\sin\theta\cos\theta + 1 - \sin^2\theta = -\sin\theta\cos\theta + \cos^2\theta = \cos\theta(\cos\theta - \sin\theta) = 0$

So $\cos\theta = 0$ or $\tan\theta = 1$ but $\cos\theta \neq 0$

$\therefore \theta = -\frac{\pi}{4}$ or $\theta = -\frac{3\pi}{4}$

When $\theta = -\frac{\pi}{4}$, $\quad x = \sqrt{2}$

When $\theta = -\frac{3\pi}{4}$, $\quad x = -\sqrt{2}$

So the tangent lines are $x = \sqrt{2}$ and $x = -\sqrt{2}$

6) a)

Tangent parallel to the initial line $\theta = 0$: $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0$

$r = 2 + \sqrt{3}\cos\theta \Rightarrow r\sin\theta = 2\sin\theta + \sqrt{3}\sin\theta\cos\theta \Rightarrow y = 2\sin\theta + \sqrt{3}\sin\theta\cos\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = 2\cos\theta - \sqrt{3}\sin^2\theta + \sqrt{3}\cos^2\theta$

Consider $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$:

$2\cos\phi - \sqrt{3}\sin^2\phi + \sqrt{3}\cos^2\phi = 2\cos\phi - \sqrt{3}(1 - \cos^2\phi) + \sqrt{3}\cos^2\phi$

$\therefore 2\sqrt{3}\cos^2\phi + 2\cos\phi - \sqrt{3} = 0$

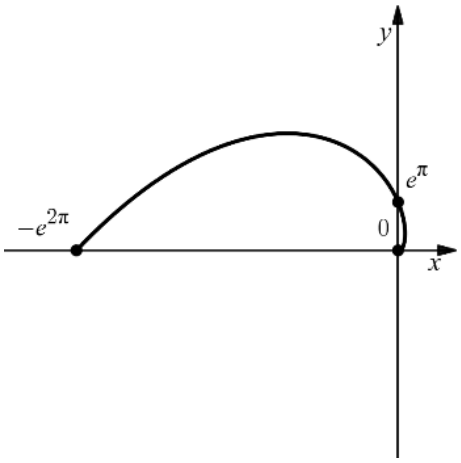
So $\cos\phi = \frac{-2\pm\sqrt{4-(4\times2\sqrt{3}\times-\sqrt{3})}}{4\sqrt{3}} = \frac{-2\pm\sqrt{28}}{4\sqrt{3}} = \frac{-2\pm2\sqrt{7}}{4\sqrt{3}} = \frac{-1\pm\sqrt{7}}{2\sqrt{3}}$

Since $0 < \phi < \frac{\pi}{2}$ then $\cos\phi > 0$ so $r = \frac{-1+\sqrt{7}}{2\sqrt{3}}$ as req.

6) b)

$r = 2 + \sqrt{3}\left(\frac{-1+\sqrt{7}}{2\sqrt{3}}\right) = 2 + \frac{-1+\sqrt{7}}{2} = \frac{3+\sqrt{7}}{2}$

7) a)



7) b)

$e^{2\pi}$

7) c)

Tangent perpendicular to the initial line $\theta = 0$: $\frac{dx}{d\theta} = 0$ and $\frac{dy}{d\theta} \neq 0$

$r = e^{2\theta} \Rightarrow r\cos\theta = e^{2\theta}\cos\theta \Rightarrow x = e^{2\theta}\cos\theta$

$\frac{dx}{d\theta} = -e^{2\theta}\sin\theta + 2e^{2\theta}\cos\theta$

Consider $\frac{dx}{d\theta} = 0$:

$-e^{2\theta}\sin\theta + 2e^{2\theta}\cos\theta = e^{2\theta}(2\cos\theta - \sin\theta) = 0$

So $2\cos\theta - \sin\theta = 0 \Rightarrow 2\cos\theta = \sin\theta$

So $\tan\theta = 2$

$\therefore \theta = \arctan(2)$

So that $r = e^{2\arctan(2)}$ so the distance is $e^{2\arctan(2)} = 9.15$ units (3 SF)

7) d)

Tangent parallel to the initial line $\theta = 0$: $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0$

$r = \mathrm{e}^{2\theta} \Rightarrow r\sin\theta = \mathrm{e}^{2\theta}\sin\theta \Rightarrow y = \mathrm{e}^{2\theta}\sin\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = \mathrm{e}^{2\theta}\cos\theta + 2\mathrm{e}^{2\theta}\sin\theta$

Consider $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$:

$\mathrm{e}^{2\theta}\cos\theta + 2\mathrm{e}^{2\theta}\sin\theta = \mathrm{e}^{2\theta}(\cos\theta + 2\sin\theta) = 0$

So $\cos\theta + 2\sin\theta = 0 \Rightarrow 2\sin\theta = -\cos\theta$

So $\tan\theta = -\frac{1}{2}$ for $0 < \theta < \pi$

$\therefore \theta = \pi - \arctan\left(\frac{1}{2}\right)$

So that $r = \mathrm{e}^{2\left(\pi - \arctan\left(\frac{1}{2}\right)\right)}$ so the distance is $\mathrm{e}^{2\left(\pi - \arctan\left(\frac{1}{2}\right)\right)} = 212$ units (3 SF)

8)

Find position of line AD:

Tangent perpendicular to the initial line $\theta = 0$: $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}y}{\mathrm{d}\theta} \neq 0$

$r = \frac{1}{2}\sin 2\theta \Rightarrow r\cos\theta = \frac{1}{2}\sin 2\theta\cos\theta = \sin\theta\cos^2\theta \Rightarrow x = \sin\theta\cos^2\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -2\sin^2\theta\cos\theta + \cos^3\theta$

Consider $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$:

$-2\sin^2\theta\cos\theta + \cos^3\theta = \cos\theta(\cos^2\theta - 2\sin^2\theta)$

$= 0$

$\cos^2\theta - 2\sin^2\theta = 0 \Rightarrow 2\tan^2\theta = 1 \Rightarrow \tan\theta = \pm\frac{1}{\sqrt{2}}$

So $\cos\theta = 0$ or $\tan\theta = \pm\frac{1}{\sqrt{2}}$

Consider $\tan\theta = \frac{1}{\sqrt{2}}$ so $\sin\theta = \frac{1}{\sqrt{3}}$ and $\cos\theta = \sqrt{\frac{2}{3}}$

So that $x = \frac{1}{\sqrt{3}} \times \frac{2}{3} = \frac{2\sqrt{3}}{9}$

Therefore the area of the square is $\left(2 \times \frac{2\sqrt{3}}{9}\right)^2 = \frac{16}{27}$ units^2

9) a)

Tangent parallel to the initial line $\theta = 0$: $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0$

$r = a + a\cos\theta \Rightarrow r\sin\theta = a\sin\theta + a\sin\theta\cos\theta \Rightarrow y = a\sin\theta + \frac{1}{2}a\sin 2\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = a\cos\theta + a\cos 2\theta$

Consider $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$:

$a\cos\theta + a\cos 2\theta = a\cos\theta + a(2\cos^2\theta - 1) = 2a\cos^2\theta + a\cos\theta - a$

$= a(\cos\theta + 1)(2\cos\theta - 1)$

$= 0$

So $\cos\theta = -1$ or $\cos\theta = \frac{1}{2}$

$\therefore \theta = \pi$, $\theta = \frac{\pi}{3}$ or $\theta = \frac{5\pi}{3}$

When $\theta = \pi$, $r = a + a\cos\theta = 0$

When $\theta = \frac{\pi}{3}$, $r = \frac{3}{2}a$

When $\theta = \frac{5\pi}{3}$, $y = \frac{3}{2}a$

So $P\left(\frac{3}{2}a, \frac{\pi}{3}\right)$ and $Q\left(\frac{3}{2}a, \frac{5\pi}{3}\right)$

9) b)

Symmetrical, so consider when $\theta = \frac{\pi}{3}$, then $y = \frac{3\sqrt{3}}{4}a$

The distance is $2 \times \frac{3\sqrt{3}}{4}a = 12 \therefore a = \frac{8}{\sqrt{3}} = \frac{8\sqrt{3}}{3}$

10) a)

Tangent perpendicular to the initial line $\theta = 0$: $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}y}{\mathrm{d}\theta} \neq 0$

$r = a - a\sin\theta \Rightarrow r\cos\theta = a\cos\theta - a\sin\theta\cos\theta \Rightarrow x = a\cos\theta - \frac{1}{2}a\sin 2\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -a\sin\theta - a\cos 2\theta$

Consider $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$:

$-a\sin\theta - a\cos 2\theta = -a\sin\theta - a(1 - 2\sin^2\theta) = 2a\sin^2\theta - a\sin\theta - a$

$= a(\sin\theta - 1)(2\sin\theta + 1)$

$= 0$

So $\sin\theta = 1$ or $\sin\theta = -\frac{1}{2}$

$\therefore \theta = 0, \theta = \pi, \theta = \frac{7\pi}{6}$ or $\theta = \frac{11\pi}{6}$

When $\theta = 0, \pi$ $\quad r = a - a\sin\theta = 0$

When $\theta = \frac{7\pi}{6}$, $\quad r = \frac{3}{2}a$

When $\theta = \frac{11\pi}{6}$, $\quad r = \frac{3}{2}a$

So $P\left(\frac{3}{2}a, \frac{7\pi}{6}\right)$ and $Q\left(\frac{3}{2}a, \frac{11\pi}{6}\right)$

10) b)

Symmetrical, so consider when $\theta = \frac{7\pi}{6}$, then $x = -\frac{3\sqrt{3}}{4}a$

The distance is $2 \times \frac{3\sqrt{3}}{4}a = 5 \therefore a = \frac{10}{3\sqrt{3}} = \frac{10\sqrt{3}}{9}$

11) a)

Tangent perpendicular to the initial line $\theta = 0$: $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}y}{\mathrm{d}\theta} \neq 0$

$r = k - 3\cos\theta \Rightarrow r\cos\theta = k\cos\theta - 3\cos^2\theta \Rightarrow x = k\cos\theta - 3\cos^2\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -k\sin\theta + 6\cos\theta\sin\theta$

Consider $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$:

$-k\sin\theta + 6\cos\theta\sin\theta = \sin\theta(6\cos\theta - k) = 0$

So $\sin\theta = 0$ or $\cos\theta = \frac{k}{6}$

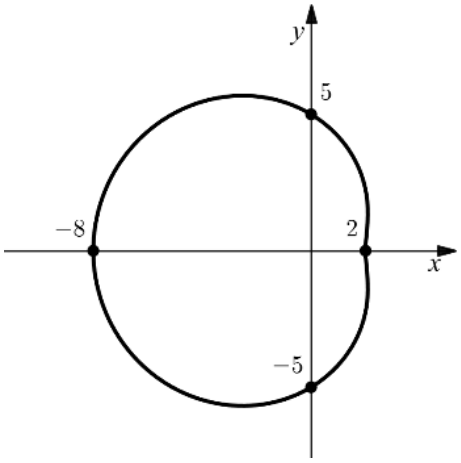
For $\sin\theta = 0$ we obtain two of the points at $\theta = 0$ and $\theta = \pi$

The other two points come from solving $\cos\theta = \frac{k}{6}$ so it must be that $-1 < \frac{k}{6} < 1$

That is, $-6 < k < 6$

But since $k > 3$ then $3 < k < 6$ as req.

11) b)



12) a)

Tangent parallel to the initial line $\theta = 0$: $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0$

$r = k - 4\sin\theta \Rightarrow r\sin\theta = k\sin\theta - 4\sin^2\theta \Rightarrow y = k\sin\theta - 4\sin^2\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = k\cos\theta - 8\cos\theta\sin\theta$

Consider $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$:

$k\cos\theta - 8\cos\theta\sin\theta = \cos\theta\,(k - 8\sin\theta) = 0$

So $\cos\theta = 0$ or $\sin\theta = \frac{k}{8}$

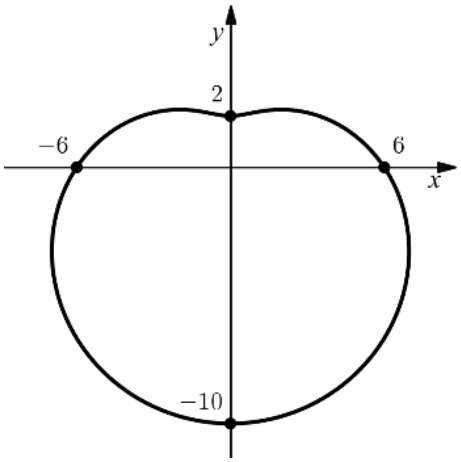
For $\cos\theta = 0$ we obtain two of the points at $\theta = \frac{\pi}{2}$ and $\theta = \frac{3\pi}{2}$

The other two points come from solving $\sin\theta = \frac{k}{8}$ so it must be that $-1 < \frac{k}{8} < 1$

That is, $-8 < k < 8$

But since $k > 4$ then $4 < k < 8$ as req.

12) b)



13) a)

Tangent parallel to the initial line $\theta = 0$: $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}x}{\mathrm{d}\theta} \neq 0$

$r = 4 - 4\sin\theta \Rightarrow r\sin\theta = 4\sin\theta - 4\sin^2\theta \Rightarrow y = 4\sin\theta - 4\sin^2\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = 4\cos\theta - 8\sin\theta\cos\theta$

$r = 4 - 4\sin\theta \Rightarrow r\cos\theta = 4\cos\theta - 4\sin\theta\cos\theta \Rightarrow x = 4\cos\theta - 2\sin 2\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -4\sin\theta - 4\cos 2\theta$

Consider $\frac{\mathrm{d}y}{\mathrm{d}\theta} = 0$:

$4\cos\theta - 8\sin\theta\cos\theta = 4\cos\theta\,(1 - 2\sin\theta) = 0$

So $\cos\theta = 0$ or $\sin\theta = \frac{1}{2}$

$\therefore \theta = \frac{\pi}{2}, \theta = \frac{\pi}{6}, \theta = \frac{5\pi}{6}$

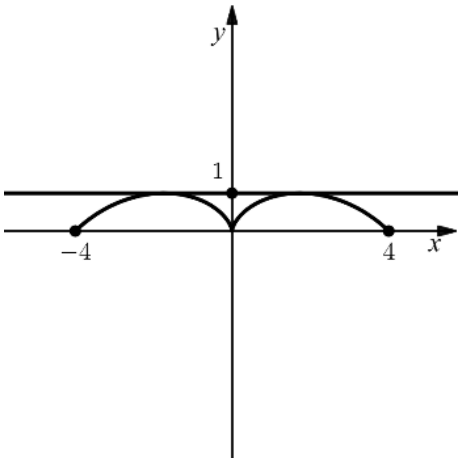
When $\theta = \frac{\pi}{2}$, $y = 0$ and but $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ so a cusp point.

When $\theta = \frac{\pi}{6}$, $y = 1$

When $\theta = \frac{5\pi}{6}$, $y = 1$

So the tangent line L is $y = 1$

13) b)



13) c)

The line L and the curve C intersect when $\theta = \frac{\pi}{6}$ and $\theta = \frac{5\pi}{6}$

When $\theta = \frac{\pi}{6}$, $x = 4\cos\frac{\pi}{6} - 2\sin\frac{\pi}{3} = \sqrt{3}$

So area of rectangle under the line L is $1 \times 2\sqrt{3} = 2\sqrt{3}$

Now area under polar curve:

$$\frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (4 - 4\sin\theta)^2 \,\mathrm{d}\theta = 8\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (1 - \sin\theta)^2 \,\mathrm{d}\theta = 8\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (1 - 2\sin\theta + \sin^2\theta) \,\mathrm{d}\theta$$

$$= 8\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} 1 \,\mathrm{d}\theta - 16\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \sin\theta \,\mathrm{d}\theta + 8\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \sin^2\theta \,\mathrm{d}\theta$$

$$= 8\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} 1 \,\mathrm{d}\theta - 16\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} \sin\theta \,\mathrm{d}\theta + 4\int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} (1 - \cos 2\theta) \,\mathrm{d}\theta$$

$$= 8[\theta]_{\frac{\pi}{6}}^{\frac{5\pi}{6}} + 16[\cos\theta]_{\frac{\pi}{6}}^{\frac{5\pi}{6}} + 4\left[\theta - \frac{1}{2}\sin 2\theta\right]_{\frac{\pi}{6}}^{\frac{5\pi}{6}}$$

$$= 8\left[\frac{5\pi}{6} - \frac{\pi}{6}\right] + 16\left[-\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}\right] + 4\left[\frac{5\pi}{6} + \frac{\sqrt{3}}{4} - \frac{\pi}{6} + \frac{\sqrt{3}}{4}\right]$$

$$= \frac{16\pi}{3} - 16\sqrt{3} + 2\sqrt{3} + \frac{8\pi}{3}$$

$$= 8\pi - 14\sqrt{3}$$

The area of the triangle between $(\sqrt{3}, 1)$, $(-\sqrt{3}, 1)$ and $(0, 0)$ is given by

$$\frac{1}{2} \times \sqrt{(\sqrt{3})^2 + 1} \times \sqrt{(\sqrt{3})^2 + 1} \times \sin\left(\frac{5\pi}{6} - \frac{\pi}{6}\right) = \sqrt{3}$$

So the area wanted is given by the triangle between points:

$$\sqrt{3} - (8\pi - 14\sqrt{3}) = (15\sqrt{3} - 8\pi) \text{ units}^2$$

14) a)

Tangent perpendicular to the initial line $\theta = 0$: $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}y}{\mathrm{d}\theta} \neq 0$

$r = 2 + \tan\theta \Rightarrow r\cos\theta = 2\cos\theta + \cos\theta\tan\theta \Rightarrow x = 2\cos\theta + \cos\theta\tan\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -2\sin\theta + \cos\theta\sec^2\theta - \sin\theta\tan\theta$

$r = 2 + \tan\theta \Rightarrow r\sin\theta = 2\sin\theta + \sin\theta\tan\theta \Rightarrow y = 2\sin\theta + \sin\theta\tan\theta$

$\frac{\mathrm{d}y}{\mathrm{d}\theta} = 2\cos\theta + \sin\theta\sec^2\theta + \cos\theta\tan\theta$

Consider $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$:

$-2\sin\theta + \cos\theta\sec^2\theta - \sin\theta\tan\theta = 0$

So $-2\sin\theta + \frac{1}{\cos\theta} - \frac{\sin^2\theta}{\cos\theta} = 0$

$\Rightarrow -2\sin\theta\cos\theta + 1 - \sin^2\theta = -2\sin\theta\cos\theta + \cos^2\theta = \cos\theta(\cos\theta - 2\sin\theta) = 0$

So $\cos\theta = 0$ or $\tan\theta = \frac{1}{2}$ but $\cos\theta \neq 0$

$\therefore \tan\theta = \frac{1}{2}$ (and the angle is acute) so that $\sin\theta = \frac{1}{\sqrt{5}}$ (and $\cos\theta = \frac{2}{\sqrt{5}}$) as req.

14) b)

$x = 2\cos\theta + \cos\theta\tan\theta = 2 \times \frac{2}{\sqrt{5}} + \frac{2}{\sqrt{5}} \times \frac{1}{2} = \sqrt{5}$

14) c)

$r = 2 + \tan\theta = 2 + \frac{1}{2} = \frac{5}{2}$

Area of triangle using $\frac{1}{2}ab\sin C$: $\frac{1}{2} \times \frac{5}{2} \times \sqrt{5} \times \frac{1}{\sqrt{5}} = \frac{5}{4}$

Now area under polar curve:

$\frac{1}{2}\int(2 + \tan\theta)^2 \, \mathrm{d}\theta = \frac{1}{2}\int(4 + 4\tan\theta + \tan^2\theta) \, \mathrm{d}\theta$

$= \frac{1}{2}\int 4 \, \mathrm{d}\theta + \frac{1}{2}\int 4\tan\theta \, \mathrm{d}\theta + \frac{1}{2}\int \tan^2\theta \, \mathrm{d}\theta$

$= 2\int 1 \, \mathrm{d}\theta + 2\int \tan\theta \, \mathrm{d}\theta + \frac{1}{2}\int(\sec^2\theta - 1) \, \mathrm{d}\theta$

$= 2\int 1 \, \mathrm{d}\theta + 2\int \tan\theta \, \mathrm{d}\theta + \frac{1}{2}\int \sec^2\theta \, \mathrm{d}\theta - \frac{1}{2}\int 1 \, \mathrm{d}\theta$

$= 2\theta + 2\ln|\sec\theta| + \frac{1}{2}\tan\theta - \frac{1}{2}\theta \; (+c)$

$= \frac{3}{2}\theta + 2\ln|\sec\theta| + \frac{1}{2}\tan\theta \; (+c)$

So that where $\phi = \arctan\frac{1}{2}$ we have:

$\frac{1}{2}\int_0^{\phi}(2 + \tan\theta)^2 \, \mathrm{d}\theta = \left(\frac{3}{2}\phi + 2\ln|\sec\phi| + \frac{1}{2}\tan\phi\right) - \left(0 + 2\ln|\sec 0| + \frac{1}{2}\tan 0\right)$

$= \left(\frac{3}{2}\phi + 2\ln|\sec\phi| + \frac{1}{2}\tan\phi\right) - (0 + 0 + 0)$

Given that $\sec\phi = \frac{\sqrt{5}}{2}$ and $\tan\phi = \frac{1}{2}$:

$= \frac{3}{2}\arctan\left(\frac{1}{2}\right) + 2\ln\left|\frac{\sqrt{5}}{2}\right| + \frac{1}{4}$

So the area wanted is:

$\frac{5}{4} - \left(\frac{3}{2}\arctan\left(\frac{1}{2}\right) + 2\ln\left|\frac{\sqrt{5}}{2}\right| + \frac{1}{4}\right) = 0.0814$ units^2 (3 SF)

15) a)

Tangent perpendicular to the initial line $\theta = 0$: $\frac{\mathrm{d}x}{\mathrm{d}\theta} = 0$ and $\frac{\mathrm{d}y}{\mathrm{d}\theta} \neq 0$

$r = 6\sqrt{3} - 2\cos 3\theta \Rightarrow r\cos\theta = 6\sqrt{3}\cos\theta - 2\cos\theta\cos 3\theta$

$\Rightarrow x = 6\sqrt{3}\cos\theta - 2\cos\theta\cos 3\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -6\sqrt{3}\sin\theta + 2\sin\theta\cos 3\theta + 6\cos\theta\sin 3\theta$

When $\theta = \frac{\pi}{6}$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -6\sqrt{3}\sin\frac{\pi}{6} + 2\sin\frac{\pi}{6}\cos\frac{\pi}{2} + 6\cos\frac{\pi}{6}\sin\frac{\pi}{2}$

$= -3\sqrt{3} + 0 + 3\sqrt{3} = 0$ as req.

15) b)

Find area above axes and double:

$r = 6\sqrt{3} - 2\cos 3\theta = 6\sqrt{3} - 0 = 6\sqrt{3}$

When $\theta = \frac{\pi}{6}$, $x = 6\sqrt{3}\cos\frac{\pi}{6} - 2\cos\frac{\pi}{6}\cos\frac{\pi}{2} = 9$

Area of triangle using $\frac{1}{2}ab\sin C$: $\frac{1}{2} \times 6\sqrt{3} \times 9 \times \sin\frac{\pi}{6} = \frac{27\sqrt{3}}{2}$

Now area under polar curve:

$\frac{1}{2}\int\left(6\sqrt{3} - 2\cos 3\theta\right)^2 \mathrm{d}\theta = \frac{1}{2}\int\left(108 - 24\sqrt{3}\cos 3\theta + 4\cos^2 3\theta\right)\mathrm{d}\theta$

$= \frac{1}{2}\int 108\,\mathrm{d}\theta - 12\sqrt{3}\int\cos 3\theta\,\mathrm{d}\theta + \int 2\cos^2 3\theta\,\mathrm{d}\theta$

$= 54\int 1\,\mathrm{d}\theta - 12\sqrt{3}\int\cos 3\theta\,\mathrm{d}\theta + \int(1 + \cos 6\theta)\,\mathrm{d}\theta$

$= 54\theta - 12\sqrt{3} \times \frac{1}{3}\sin 3\theta + \theta + \frac{1}{6}\sin 6\theta\ (+c)$

$= 54\theta - 4\sqrt{3}\sin 3\theta + \theta + \frac{1}{6}\sin 6\theta\ (+c)$

So we have:

$\frac{1}{2}\int_0^{\frac{\pi}{6}}\left(6\sqrt{3} - 2\cos 3\theta\right)^2 \mathrm{d}\theta$

$= \left(54 \times \frac{\pi}{6} - 4\sqrt{3}\sin\frac{\pi}{2} + \frac{\pi}{6} + \frac{1}{6}\sin\pi\right) - \left(0 - 6\sqrt{3}\sin 0 + 0 + \frac{1}{6}\sin 0\right)$

$= \left(9\pi - 4\sqrt{3} + \frac{\pi}{6} + 0\right) - (0 - 0 + 0 + 0)$

$= \frac{55\pi}{6} - 4\sqrt{3}$

So triangle – curved area:

$\frac{27\sqrt{3}}{2} - \left(\frac{55\pi}{6} - 4\sqrt{3}\right) = \frac{35\sqrt{3}}{2} - \frac{55\pi}{6}$

Double this area:

$\left(35\sqrt{3} - \frac{55\pi}{3}\right)$ units^2

16) a)

$r = 3\cos\theta$ so $\frac{\mathrm{d}r}{\mathrm{d}\theta} = -3\sin\theta$

Maximum r is when $\frac{\mathrm{d}r}{\mathrm{d}\theta} = 0$ so $-3\sin\theta = 0$ so for $0 \le \theta \le \frac{\pi}{2}$ this is when $\theta = 0$

So $r = 3\cos 0 = 3$

Interval endpoints: $r = 3$ and $r = 0$

Maximum is $r = 3$ when $\theta = 0$

16) b)

$r = 2 + \sin\theta$ so $\frac{\mathrm{d}r}{\mathrm{d}\theta} = \cos\theta$

Maximum r is when $\frac{\mathrm{d}r}{\mathrm{d}\theta} = 0$ so $\cos\theta = 0$ so for $0 \leq \theta \leq \pi$ this is when $\theta = \frac{\pi}{2}$

So $r = 2 + \sin\frac{\pi}{2} = 3$

Interval endpoints: $r = 2$ and $r = 3$

Maximum is $r = 3$ when $\theta = \frac{\pi}{2}$

16) c)

$r = 5\cos 2\theta$ so $\frac{\mathrm{d}r}{\mathrm{d}\theta} = -10\sin 2\theta$

Maximum r is when $\frac{\mathrm{d}r}{\mathrm{d}\theta} = 0$ so $-10\sin 2\theta = 0$ so for $0 \leq \theta \leq \frac{\pi}{4}$ this is when $\theta = 0$

So $r = 5\cos 0 = 5$

Interval endpoints: $r = 5$ and $r = 0$

Maximum is $r = 5$ when $\theta = 0$

16) d)

$r = \ln(1 + \sin\theta)$ so $\frac{\mathrm{d}r}{\mathrm{d}\theta} = \frac{\cos\theta}{1+\sin\theta}$

Maximum r is when $\frac{\mathrm{d}r}{\mathrm{d}\theta} = 0$ so $\cos\theta = 0$ so for $0 \leq \theta < \pi$ this is when $\theta = \frac{\pi}{2}$

So $r = \ln\left(1 + \sin\frac{\pi}{2}\right) = \ln 2$

Interval endpoints: $r = \ln 1 = 0$ and $r = \ln 1 = 0$

Maximum is $r = \ln 2$ when $\theta = \frac{\pi}{2}$

17) a)

$r = 4\cos(2\theta)\cosh\left(\frac{\theta}{2}\right)$

When $\theta = \frac{\pi}{4}$, $r = 4\cos\left(\frac{\pi}{2}\right)\cosh\left(\frac{\pi}{8}\right) = 0$ and when $\theta = \frac{3\pi}{4}$, $r = 4\cos\left(\frac{3\pi}{2}\right)\cosh\left(\frac{3\pi}{8}\right) = 0$

And since $\cos 2\theta = 0$ only at $\frac{\pi}{4}$ and $\frac{3\pi}{4}$ it starts and ends at the pole.

On the interval $\frac{\pi}{4} < \theta < \frac{3\pi}{4}$ we know that $\cos 2\theta < 0$ and $\cosh\left(\frac{\theta}{2}\right) > 0$ so $r < 0$ for the whole interval, and so the curve consists of only one loop.

17) b)

$r = 4\cos(2\theta)\cosh\left(\frac{\theta}{2}\right)$ and so $\frac{\mathrm{d}r}{\mathrm{d}\theta} = 2\cos(2\theta)\sinh\left(\frac{\theta}{2}\right) - 8\sin(2\theta)\cosh\left(\frac{\theta}{2}\right)$

Interval endpoints: $r = 0$ and $r = 0$

Maximum r is when $\frac{\mathrm{d}r}{\mathrm{d}\theta} = 0$ at $\theta = \phi$

$2\cos(2\phi)\sinh\left(\frac{\phi}{2}\right) - 8\sin(2\phi)\cosh\left(\frac{\phi}{2}\right) = 0$

$\cos(2\phi)\sinh\left(\frac{\phi}{2}\right) = 4\sin(2\phi)\cosh\left(\frac{\phi}{2}\right)$

$\frac{\sinh\left(\frac{\phi}{2}\right)}{\cosh\left(\frac{\phi}{2}\right)} = 4\frac{\sin(2\phi)}{\cos(2\phi)}$

$\frac{\frac{1}{2}\left(\mathrm{e}^{\frac{\phi}{2}} - \mathrm{e}^{-\frac{\phi}{2}}\right)}{\frac{1}{2}\left(\mathrm{e}^{\frac{\phi}{2}} + \mathrm{e}^{-\frac{\phi}{2}}\right)} = 4\tan 2\phi$

$\mathrm{e}^{\frac{\phi}{2}} - \mathrm{e}^{-\frac{\phi}{2}} = 4\tan 2\phi\left(\mathrm{e}^{\frac{\phi}{2}} + \mathrm{e}^{-\frac{\phi}{2}}\right)$

$\mathrm{e}^{\frac{\phi}{2}} - \mathrm{e}^{-\frac{\phi}{2}} = 4\mathrm{e}^{\frac{\phi}{2}}\tan 2\phi + 4\mathrm{e}^{-\frac{\phi}{2}}\tan 2\phi$

$\mathrm{e}^{\frac{\phi}{2}} - 4\mathrm{e}^{\frac{\phi}{2}}\tan 2\phi = \mathrm{e}^{-\frac{\phi}{2}} + 4\mathrm{e}^{-\frac{\phi}{2}}\tan 2\phi$

$\mathrm{e}^{\frac{\phi}{2}}(1 - 4\tan 2\phi) = \mathrm{e}^{-\frac{\phi}{2}}(1 + 4\tan 2\phi)$

$\mathrm{e}^{\phi} = \frac{1+4\tan 2\phi}{1-4\tan 2\phi}$

$\phi = \ln\left(\frac{1+4\tan 2\phi}{1-4\tan 2\phi}\right)$ as req.

17) c)

When $\phi = 1.65488$, $r = 4\cos(2 \times 1.65488)\cosh\left(\frac{1.65488}{2}\right) = -5.372$ (4 SF)

So the distance is 5.372 units

18)

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = -r\sin\theta + \frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta$, $\frac{\mathrm{d}y}{\mathrm{d}\theta} = r\cos\theta + \frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta$

So that

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\frac{\mathrm{d}y}{\mathrm{d}\theta}}{\frac{\mathrm{d}x}{\mathrm{d}\theta}} = \frac{r\cos\theta + \frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta}{-r\sin\theta + \frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta} = \frac{\frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta}{\frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta}$ as req.

19) a)

$r = 2\theta$ so $\frac{\mathrm{d}r}{\mathrm{d}\theta} = 2$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta}{\frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta} = \frac{2\sin\theta + 2\theta\cos\theta}{2\cos\theta - 2\theta\sin\theta}$$

So

$$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{\theta=\frac{\pi}{2}} = \frac{2\sin\frac{\pi}{2} + \pi\cos\frac{\pi}{2}}{2\cos\frac{\pi}{2} - \pi\sin\frac{\pi}{2}} = -\frac{2}{\pi}$$

19) b)

$r = 2\cos\theta$ so $\frac{\mathrm{d}r}{\mathrm{d}\theta} = -2\sin\theta$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta}{\frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta} = \frac{-2\sin\theta\sin\theta + 2\cos\theta\cos\theta}{-2\sin\theta\cos\theta - 2\cos\theta\sin\theta}$$

So

$$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{\theta=\frac{\pi}{6}} = \frac{-2\sin\frac{\pi}{6}\sin\frac{\pi}{6} + 2\cos\frac{\pi}{6}\cos\frac{\pi}{6}}{-2\sin\frac{\pi}{6}\cos\frac{\pi}{6} - 2\cos\frac{\pi}{6}\sin\frac{\pi}{6}} = \frac{-\frac{1}{2}+\frac{3}{2}}{-\frac{\sqrt{3}}{2}-\frac{\sqrt{3}}{2}} = \frac{-\sqrt{3}}{3}$$

19) c)

$r = 1 + 2\sin\theta$ so $\frac{\mathrm{d}r}{\mathrm{d}\theta} = 2\cos\theta$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta}{\frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta} = \frac{2\cos\theta\sin\theta + (1+2\sin\theta)\cos\theta}{2\cos\theta\cos\theta - (1+2\sin\theta)\sin\theta}$$

So

$$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{\theta=\frac{3\pi}{4}} = \frac{2\cos\frac{3\pi}{4}\sin\frac{3\pi}{4} + \left(1+2\sin\frac{3\pi}{4}\right)\cos\frac{3\pi}{4}}{2\cos\frac{3\pi}{4}\cos\frac{3\pi}{4} - \left(1+2\sin\frac{3\pi}{4}\right)\sin\frac{3\pi}{4}} = \frac{-1-1-\frac{\sqrt{2}}{2}}{1-1-\frac{\sqrt{2}}{2}} = \frac{2+\frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2}} = 1 + 2\sqrt{2}$$

20) a)

$r = \theta \sin\theta$ and so $\frac{\mathrm{d}r}{\mathrm{d}\theta} = \theta\cos\theta + \sin\theta$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{(\theta\cos\theta+\sin\theta)\sin\theta+\theta\sin\theta\cos\theta}{(\theta\cos\theta+\sin\theta)\cos\theta-\theta\sin\theta\sin\theta}$

Expanding brackets and collecting like terms:

$= \frac{\theta\cos\theta\sin\theta+\sin^2\theta+\theta\sin\theta\cos\theta}{\theta\cos^2\theta+\sin\theta\cos\theta-\theta\sin^2\theta}$

$= \frac{2\theta\cos\theta\sin\theta+\sin^2\theta}{\theta(\cos^2\theta-\sin^2\theta)+\sin\theta\cos\theta}$

Using $\cos^2\theta - \sin^2\theta = \cos 2\theta$:

$= \frac{2\theta\cos\theta\sin\theta+\sin^2\theta}{\theta\cos 2\theta+\sin\theta\cos\theta}$

Multiplying numerator and denominator by 2:

$= \frac{4\theta\cos\theta\sin\theta+2\sin^2\theta}{2\theta\cos 2\theta+2\sin\theta\cos\theta}$

Using $2\cos\theta\sin\theta = \sin 2\theta$:

$= \frac{2\theta\sin 2\theta+2\sin^2\theta}{2\theta\cos 2\theta+\sin 2\theta}$ as req.

20) b)

Solving the numerator set to zero:

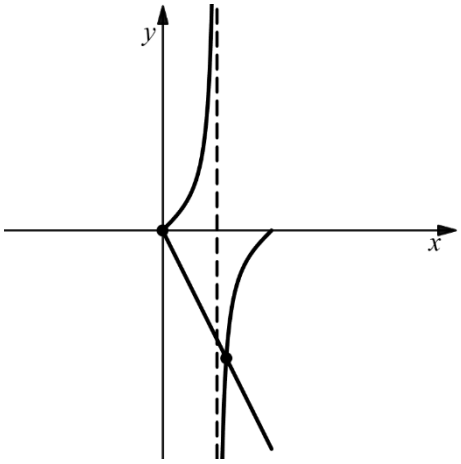
$4\theta \cos\theta \sin\theta + 2\sin^2\theta = 0$

$\sin\theta\,(2\theta\cos\theta + \sin\theta) = 0$

$\sin\theta = 0$ or $2\theta\cos\theta + \sin\theta = 0 \Rightarrow \sin\theta = -2\theta\cos\theta \therefore \tan\theta = -2\theta$

For $0 \leq \theta < \pi$

$\sin\theta = 0$ has 1 solution which is when $\theta = 0$

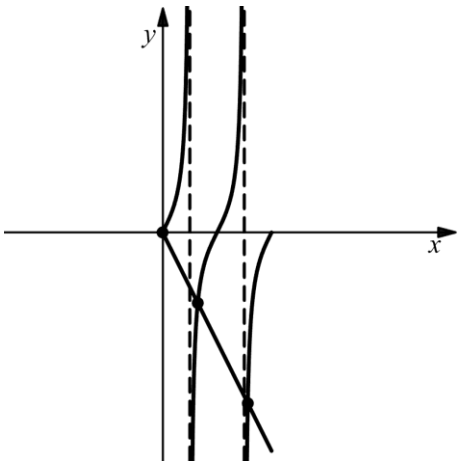


$\tan\theta = -2\theta$ has 2 solutions, one of which is when $\theta = 0$

Solving the denominator set to zero:

$2\theta\cos 2\theta + \sin 2\theta = 0 \Rightarrow \sin 2\theta = -2\theta\cos 2\theta \Rightarrow \tan 2\theta = -2\theta$

For $0 \leq \theta < \pi$



$\tan 2\theta = -2\theta$ has 3 solutions, one of which is when $\theta = 0$

So there are is one horizontal tangent and two vertical tangents.

When $\theta = 0$, $r = 0$ but also if $\theta = \pi$ then also $r = 0$ and since there is one horizontal tangent then there is one loop.

21) a)

$r = \mathrm{e}^{k\theta}$ and so $\frac{\mathrm{d}r}{\mathrm{d}\theta} = k\mathrm{e}^{k\theta}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{k\mathrm{e}^{k\theta}\sin\theta + \mathrm{e}^{k\theta}\cos\theta}{k\mathrm{e}^{k\theta}\cos\theta - \mathrm{e}^{k\theta}\sin\theta}$

Since $\mathrm{e}^{k\theta} \neq 0$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{k\sin\theta + \cos\theta}{k\cos\theta - \sin\theta}$

21) b)

The line OP makes an angle of θ with the x-axis

Let A be the point at which the tangent crosses the positive x-axis so $\phi = \widehat{OPA}$

Further let α be the obtuse angle that the tangent makes with the x-axis at the point A

By geometry: $\theta + \phi = \alpha \Rightarrow \phi = \alpha - \theta$

$\tan\phi = \tan(\alpha - \theta) = \frac{\tan\alpha - \tan\theta}{1 + \tan\alpha\tan\theta}$

But since $\tan\alpha = \frac{k\sin\theta + \cos\theta}{k\cos\theta - \sin\theta}$ we have:

$$\frac{\tan\alpha - \tan\theta}{1 + \tan\alpha\tan\theta} = \frac{\frac{k\sin\theta + \cos\theta}{k\cos\theta - \sin\theta} - \frac{\sin\theta}{\cos\theta}}{1 + \left(\frac{k\sin\theta + \cos\theta}{k\cos\theta - \sin\theta}\right)\frac{\sin\theta}{\cos\theta}}$$

Consider the numerator:

$$\frac{k\sin\theta + \cos\theta}{k\cos\theta - \sin\theta} - \frac{\sin\theta}{\cos\theta} = \frac{\cos\theta(k\sin\theta + \cos\theta) - \sin\theta(k\cos\theta - \sin\theta)}{\cos\theta(k\cos\theta - \sin\theta)}$$

$$= \frac{k\sin\theta\cos\theta + \cos^2\theta - k\cos\theta\sin\theta + \sin^2\theta}{\cos\theta(k\cos\theta - \sin\theta)}$$

$$= \frac{1}{\cos\theta(k\cos\theta - \sin\theta)}$$

Consider the denominator:

$$1 + \left(\frac{k\sin\theta + \cos\theta}{k\cos\theta - \sin\theta}\right)\frac{\sin\theta}{\cos\theta} = \frac{k\cos^2\theta - \cos\theta\sin\theta}{k\cos^2\theta - \cos\theta\sin\theta} + \frac{k\sin^2\theta + \sin\theta\cos\theta}{k\cos^2\theta - \cos\theta\sin\theta}$$

$$= \frac{k\cos^2\theta - \cos\theta\sin\theta + k\sin^2\theta + \sin\theta\cos\theta}{k\cos^2\theta - \cos\theta\sin\theta}$$

$$= \frac{k}{\cos\theta(k\cos\theta - \sin\theta)}$$

So that

$$\frac{\tan\alpha - \tan\theta}{1 + \tan\alpha\tan\theta} = \frac{\frac{1}{\cos\theta(k\cos\theta - \sin\theta)}}{\frac{k}{\cos\theta(k\cos\theta - \sin\theta)}} = \frac{1}{k} \text{ as req.}$$

22)

$r = \sin 2\theta$ and $\frac{\mathrm{d}r}{\mathrm{d}\theta} = 2\cos 2\theta$

Consider the line $\theta = \frac{\pi}{4}$

Let $\theta_1 = \frac{\pi}{4} + \phi$ and $\theta_2 = \frac{\pi}{4} - \phi$

Consider point 1 which has angle $\theta_1 = \frac{\pi}{4} + \phi$:

So that $r_1 = \sin\left(\frac{\pi}{2} + 2\phi\right) = \sin\frac{\pi}{2}\cos 2\phi - \cos\frac{\pi}{2}\sin 2\phi = \cos 2\phi$

And $\frac{\mathrm{d}r_1}{\mathrm{d}\theta_1} = 2\cos(2\theta_1) = 2\cos\left(\frac{\pi}{2} + 2\phi\right) = 2\left(\cos\frac{\pi}{2}\cos 2\phi - \sin\frac{\pi}{2}\sin 2\phi\right) = -2\sin 2\phi$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{\theta_1} = \frac{-2\sin 2\phi \sin\theta_1 + \cos 2\phi\cos\theta_1}{-2\sin 2\phi\cos\theta_1 - \cos 2\phi\sin\theta_1}$

But since $\theta_1 = \frac{\pi}{4} + \phi$ we have

$\sin\theta_1 = \sin\left(\frac{\pi}{4} + \phi\right) = \sin\frac{\pi}{4}\cos\phi + \cos\frac{\pi}{4}\sin\phi = \frac{\sqrt{2}}{2}(\cos\phi + \sin\phi)$

And $\cos\theta_1 = \cos\left(\frac{\pi}{4} + \phi\right) = \cos\frac{\pi}{4}\cos\phi - \sin\frac{\pi}{4}\sin\phi = \frac{\sqrt{2}}{2}(\cos\phi - \sin\phi)$

So that

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{\theta_1} = \frac{-2\sin 2\phi \times \frac{\sqrt{2}}{2}(\cos\phi+\sin\phi) + \cos 2\phi \times \frac{\sqrt{2}}{2}(\cos\phi-\sin\phi)}{-2\sin 2\phi \times \frac{\sqrt{2}}{2}(\cos\phi-\sin\phi) - \cos 2\phi \times \frac{\sqrt{2}}{2}(\cos\phi+\sin\phi)}$

$= \frac{-2\sqrt{2}\sin 2\phi(\cos\phi+\sin\phi) + \sqrt{2}\cos 2\phi(\cos\phi-\sin\phi)}{-2\sqrt{2}\sin 2\phi(\cos\phi-\sin\phi) - \sqrt{2}\cos 2\phi(\cos\phi+\sin\phi)}$

$= \frac{2\sqrt{2}\sin 2\phi(\cos\phi+\sin\phi) - \sqrt{2}\cos 2\phi(\cos\phi-\sin\phi)}{2\sqrt{2}\sin 2\phi(\cos\phi-\sin\phi) + \sqrt{2}\cos 2\phi(\cos\phi+\sin\phi)}$

Similarly, Consider point 2 which has angle $\theta_2 = \frac{\pi}{4} - \phi$:

$r_2 = \sin\left(\frac{\pi}{2} - 2\phi\right) = \sin\frac{\pi}{2}\cos 2\phi + \cos\frac{\pi}{2}\sin 2\phi = \cos 2\phi$

And $\frac{\mathrm{d}r_2}{\mathrm{d}\theta_2} = 2\cos(2\theta_2) = 2\cos\left(\frac{\pi}{2} - 2\phi\right) = 2\left(\cos\frac{\pi}{2}\cos 2\phi + \sin\frac{\pi}{2}\sin 2\phi\right) = 2\sin 2\phi$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{\theta_2} = \frac{2\sin 2\phi \sin\theta_2 + \cos 2\phi\cos\theta_2}{2\sin 2\phi\cos\theta_2 - \cos 2\phi\sin\theta_2}$

But since $\theta_2 = \frac{\pi}{4} - \phi$ we have

$\sin\theta_2 = \sin\left(\frac{\pi}{4} - \phi\right) = \sin\frac{\pi}{4}\cos\phi - \cos\frac{\pi}{4}\sin\phi = \frac{\sqrt{2}}{2}(\cos\phi - \sin\phi)$

And $\cos\theta_2 = \cos\left(\frac{\pi}{4} - \phi\right) = \cos\frac{\pi}{4}\cos\phi + \sin\frac{\pi}{4}\sin\phi = \frac{\sqrt{2}}{2}(\cos\phi + \sin\phi)$

So that

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{\theta_2} = \frac{2\sin 2\phi \times \frac{\sqrt{2}}{2}(\cos\phi-\sin\phi) + \cos 2\phi \times \frac{\sqrt{2}}{2}(\cos\phi+\sin\phi)}{2\sin 2\phi \times \frac{\sqrt{2}}{2}(\cos\phi+\sin\phi) - \cos 2\phi \times \frac{\sqrt{2}}{2}(\cos\phi-\sin\phi)}$

$= \frac{2\sqrt{2}\sin 2\phi(\cos\phi-\sin\phi)+\sqrt{2}\cos 2\phi(\cos\phi+\sin\phi)}{2\sqrt{2}\sin 2\phi(\cos\phi+\sin\phi)-\sqrt{2}\cos 2\phi(\cos\phi-\sin\phi)}$

Compare the expressions:

$\left.\frac{dy}{dx}\right|_{\theta_1} = \frac{2\sqrt{2}\sin 2\phi(\cos\phi+\sin\phi)-\sqrt{2}\cos 2\phi(\cos\phi-\sin\phi)}{2\sqrt{2}\sin 2\phi(\cos\phi-\sin\phi)+\sqrt{2}\cos 2\phi(\cos\phi+\sin\phi)}$

and

$\left.\frac{dy}{dx}\right|_{\theta_2} = \frac{2\sqrt{2}\sin 2\phi(\cos\phi-\sin\phi)+\sqrt{2}\cos 2\phi(\cos\phi+\sin\phi)}{2\sqrt{2}\sin 2\phi(\cos\phi+\sin\phi)-\sqrt{2}\cos 2\phi(\cos\phi-\sin\phi)}$

Compare the gradients at $\frac{\pi}{4}+\phi$ and $\frac{\pi}{4}-\phi$:

Notice that

$\left.\frac{dy}{dx}\right|_{\theta_2} = \frac{1}{\left.\frac{dy}{dx}\right|_{\theta_1}}$

The gradients at points equidistant from $\theta = \frac{\pi}{4}$ are reciprocals, so they are reflections in the line $\theta = \frac{\pi}{4}$